

Reviewer Pack — 2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$

Entry efd166108501 · INCONCLUSIVE

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2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$

Entry ID: efd166108501 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-17 00:26:59 UTC

1. Conjecture

Field A (mathematical branch): p-adic analysis (Newton polygons over $\mathbb{Q}_2$ of integer characteristic polynomials) **Field B** (complexity object): Randomized communication complexity of DISJOINT-NESS via average-case tensor invariants on sub-blocks

Statement:

Let $M_n \in \{-1, +1\}^{2n \times 2^n}$ be the signed disjointness matrix $M_n[x, y] = (-1)^{|x \cap y|}$ where x, y are interpreted as subsets of $[n]$. For each n , draw uniformly random row set R and column set C with $|R|=|C|=k=\min(2n, 40)$, form $A := M_n[R, C]$, compute its exact integer characteristic polynomial $p(t) = \sum_{i=0}^k c_i t^i$, and define the 2-adic Newton slope count $v_2(A) :=$ number of distinct slopes in the lower convex hull of $\{(i, v_2(c_i)) : c_i \neq 0\}$ where v_2 is the 2-adic valuation ($v_2(0) := +\infty$). Conjecture: $E_{\{R, C\}}[v_2(M_n^{\wedge\{R, C\}})] = \Theta(\sqrt{n})$; moreover for every $n \geq 8$ the DISJ median exceeds the median of v_2 for an i.i.d. ± 1 matrix of the same shape by a factor ≥ 1.5 , and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale (proposer's reasoning):

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a $\mathbb{Z}_2$ -valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao's principle, into a lower bound on randomized communication. Newton polygons over $\mathbb{Q}_2$ are not preserved by polynomial extension to $F_2[t]/(t^2-t)$, evading the algebrization barrier, and the invariant is structural rather than 'large+useful on truth tables', evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing ($E[v_2] \mapsto$ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Taxonomy category: AVG_T0_WORST_CASE (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 99b1aa398e6442e0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For $n \in \{10, 14, 18, 22\}$ (30 seeds each), median $v_2(\text{DISJ}) \geq 1.5 \cdot \text{median } v_2(\text{random})$ at every n , AND OLS regression of $\log(\text{median } v_2(\text{DISJ}))$ on $\log(n)$ across these four n yields slope $\in (0.35, 0.75)$. Any single n failing the ratio, or slope outside the interval, falsifies.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.90	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.90	SAFE	SAFE
ALGEBRIZATION	SAFE	0.90	SAFE	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Newton polygon 2-adic valuation characteristic polynomial random matrix - p-adic spectrum disjointness communication complexity sign matrix - 2-adic Newton slopes submatrix Hadamard disjointness tensor

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=34.8s

5.1 Generated Python source

```
import sys
import random
import math
import itertools
from fractions import Fraction

def generate_disjointness_matrix(n, rows, cols):
    matrix = []
    for i in rows:
        row = []
        for j in cols:
            intersection = set(bin(i)[2:].zfill(n)) & set(bin(j)[2:].zfill(n))
            if intersection:
                row.append(-1)
            else:
                row.append(1)
        matrix.append(row)
    return matrix

def generate_random_matrix(k):
    return [[random.choice([-1, 1]) for _ in range(k)] for _ in range(k)]

def matrix_multiply(a, b):
    result = [[0 for _ in range(len(b[0]))] for _ in range(len(a))]
```

```

    for i in range(len(a)):
        for j in range(len(b[0])):
            for k in range(len(b)):
                result[i][j] += a[i][k] * b[k][j]
    return result

def matrix_trace(m):
    return sum(m[i][i] for i in range(len(m)))

def faddeev_leverrier(matrix):
    n = len(matrix)
    char_poly = [Fraction(1)]
    for i in range(n):
        trace = matrix_trace(matrix)
        char_poly.append(-trace / (i + 1))
        if i < n - 1:
            matrix = matrix_multiply(matrix, matrix)
            for j in range(i + 1):
                matrix[j][j] += char_poly[i - j]
    return char_poly

def v2(n):
    if n == 0:
        return float('inf')
    count = 0
    while n % 2 == 0:
        n //= 2
        count += 1
    return count

def compute_newton_slope_count(coeffs):
    points = [(i, v2(abs(c))) for i, c in enumerate(coeffs) if c != 0]
    if not points:
        return 0
    points.sort()
    hull = [points[0]]
    for point in points[1:]:
        while len(hull) >= 2 and (hull[-1][1] - hull[-2][1]) * (point[0] - hull[-1][0]) <= (point[0] - hull[-2][0]) * (hull[-1][1] - hull[-2][1]):
            hull.pop()
        hull.append(point)
    slopes = []
    for i in range(1, len(hull)):
        dx = hull[i][0] - hull[i-1][0]
        dy = hull[i][1] - hull[i-1][1]
        if dx == 0:
            slope = float('inf')
        else:
            slope = dy / dx
        slopes.append(slope)
    return len(set(slopes))

def run_trial(seed):
    random.seed(seed)
    n_values = [6, 10, 14, 18, 22]
    disj_slopes = []
    random_slopes = []

```

```

for n in n_values:
    k = min(2 * n, 40)
    rows = random.sample(range(2**n), k)
    cols = random.sample(range(2**n), k)
    disj_matrix = generate_disjointness_matrix(n, rows, cols)
    disj_char_poly = faddeev_leverrier(disj_matrix)
    disj_slope_count = compute_newton_slope_count(disj_char_poly)
    disj_slopes.append(disj_slope_count)

    random_matrix = generate_random_matrix(k)
    random_char_poly = faddeev_leverrier(random_matrix)
    random_slope_count = compute_newton_slope_count(random_char_poly)
    random_slopes.append(random_slope_count)

disj_median = sorted(disj_slopes)[len(disj_slopes) // 2]
random_median = sorted(random_slopes)[len(random_slopes) // 2]
ratio = disj_median / random_median if random_median != 0 else float('inf')

conjecture_holds = ratio >= 1.5
counterexample = "" if conjecture_holds else f"Ratio {ratio} < 1.5 for seed {seed}"

return {
    "metric_name": "median_newton_slope_count",
    "metric_value": disj_median,
    "instances_tested": len(n_values),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17,
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r["metric_value"] for r in results]
    mean = sum(metric_values) / len(metric_values)
    std = math.sqrt(sum((x - mean) ** 2 for x in metric_values) / len(metric_values))
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean} std={std} support_fraction={support_fraction}")
    else:
        failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        counterexample = next(r["counterexample"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}

```

TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
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TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje

```

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                    ~~~~~^

```

KeyError: 'seed'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio $1.0 < 1.5$ (e.g., seeds 547, 593, 631, ...), satisfying the pre-registered falsification condition | next: Re-run with corrected seed-logging, stratify by $n \in \{10, 14, 18, 22\}$, and check whether the v_2 statistic saturates at small $k = \min(2n, 40)$ — perhaps a normalized slope-density (v_2/k) would better capture any DISJ-vs-random gap.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	102326
2	preregistration	claude_max	opus	0	0	6685
3	novelty	claude_max	opus	0	0	3758
4	novelty	claude_max	opus	0	0	7218
5	test_gen	mistral	codestral-latest	0	0	316523
6	test_gen	mistral	codestral-latest	0	0	313669
7	test_gen	mistral	codestral-latest	0	0	315233
8	test_gen	mistral	codestral-latest	0	0	310539
9	judge	claude_max	opus	0	0	7843

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 1383794 ms total latency. Provider mix: {'claude_max': 5, 'mistral': 4}

(full prompt+response transcripts available in `research/audit/efd166108501.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/efd166108501.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/efd166108501.tar.gz` (if generated)